
Back to entropy

Entropy-based evaluation of base language models

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Return to the language-modeling objective

Fixed benchmarks are useful slices, not a complete measure of a base model

Perplexity stopped comparing fairly

PPL is measured *per token*, but each model divides text differently: Llama-3.2 has **128,000** vocabulary entries and Qwen-2.5 has **151,643**.

BPB divides loss by UTF-8 bytes. A byte is the same amount of text for every model, so the scores share one unit.

Fixed benchmarks have blind spots

Task suites sample specific abilities, and repeated optimization on fixed tests makes contamination and benchmark-specific tuning harder to separate from general LM quality.

Entropy sees the full predictive distribution, but it needs a fair denominator, fresh target-domain text, and a contamination audit.

Go full circle: measure entropy again—with byte normalization, dynamic target-domain evaluation, controlled adaptation, and contamination filtering.

Why not just read the scorecard?

Two models, tied on language. Nine-fold apart on the exam.

A published score cannot separate **this model is better** from **this model saw more of this**.

These two sit at **adjacent positions** in our ranking and two points apart on commonsense. On grade-school maths one scores **nine times** the other.

That gap is a fact about the two **training mixes**, not about which model is better. Both are legitimate choices — the scorecard just cannot tell you which you are buying.

Qwen-3.5-4B vs Llama-3.2-1B is wider still: 2.9% apart on BPB, 73.9 points apart on GSM8K.

	Llama-3.2-1B	Qwen-2.5-1.5B
Our ruler (BPB) ↓	0.771	0.797
HellaSwag ↑	0.658	0.678
GSM8K ↑	0.065	0.611

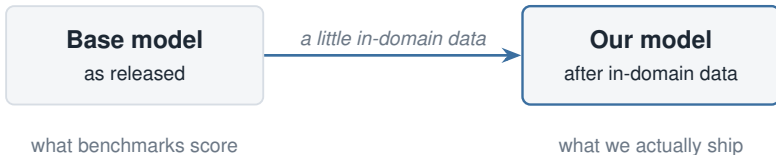
3.4% apart on language. **2.0 pts** on commonsense.
54.6 pts on maths.

And we cannot audit anyone's training data — but text written *after* they shipped cannot have been studied for.

We are not accusing anyone. A years-old public benchmark is a target whether or not anyone aims at it — so read a second axis they could not prepare for.

Measure the model we will actually use

A controlled adaptation turns evaluation into a deployment measurement



A benchmark describes the model as released. A product uses the model after domain data. We measure both states on the same held-out corpus.

The **adapted BPB level** selects the model. The reduction from zero-shot BPB explains how much calibration it needed.

Two measurements, one decision

1 — Use a fair ruler

Normalize cross-entropy onto *bytes* instead of tokens:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln 2} \times \frac{|\text{tokens}|}{|\text{bytes}|}$$

UTF-8 bytes are the same unit for every model, so BPB is **tokenizer-agnostic**. Same physical text, same denominator, comparable numbers.

Not ours — The Pile, Paloma. Under-used, and we make it the primary metric.

2 — Measure domain fit

For each model, on one identical corpus:

1. Zero-shot BPB
2. A *fixed* LoRA adaptation step — identical for every model
3. Adapted BPB

Primary output: adapted BPB for model selection.

Diagnostic: relative reduction for corpus calibration.

The pipeline

Chronologically strict: nothing is scored before it clears the gate



Corpus: one deduplicated slice of Google News, 8 June 2026 (119k articles), split 80/10/10 with zero intersection between splits. **Cohort:** 11 base models, 0.5B–35B, dense Transformers + a sparse MoE + a Liquid architecture.

Phase 1: why we trust the numbers

“Scout, then verify” — cheap detectors propose, an expensive test decides

- **Min-K++** (scout) — inspect the least-likely 20% of tokens; high confidence on rare tokens looks like recall, not prediction.
- **CoDeC** (scout, ours) — supply the preceding context; memorized text shows a non-linear *spike* in confidence.
- **Dynamic Choice Quiz** (gatekeeper) — hide the real passage among **two** perturbed variants (one word swap, one placeholder). If the model reliably picks the original, quarantine it.

Removes the passages most at risk. It is *not* proof of memorization: a model can prefer the fluent original without having seen it.

Run **per model**; we conservatively quarantine the **union** of every model's risk-flagged passages.

⇒ all 11 models are adapted and scored on **one identical, cleansed corpus**.

80 sequences removed globally.

Applies to **news**. Reddit / HN / math were scraped *after* every model shipped — clean by construction.

Results: 11 models, one ruler

Corrected fixed protocol; injected marker targets are not scored

Model	Zero-shot	Adapted
Gemma-4-31B	0.752	0.624
Gemma-4-12B	0.866	0.653
Qwen-3.5-35B-MoE	0.707	0.678
Ministral-3-14B	0.718	0.679
Qwen-3.5-9B	0.740	0.708
Qwen-2.5-7B	0.758	0.717
Qwen-3.5-4B	0.782	0.749
Llama-3.2-1B	0.811	0.771
Qwen-2.5-1.5B	0.833	0.797
LiquidAI-LFM2.5	1.340	0.852
Qwen-2.5-0.5B	0.935	0.901

Gemma-4-31B reaches the best adapted BPB: **0.624**.

The **35B MoE** lands with dense 12–14B models, giving a direct quality measurement between its active and total parameter counts.

LFM-2.5 shows the largest calibration shift: **36.4%**.

Selection uses adapted BPB.

Reduction explains how much each base distribution moved toward the corpus; it is not the ranking metric.

Two numbers, two jobs

The level chooses a model; the reduction explains calibration

Adapted BPB

The primary selection score: how efficiently the adapted model predicts held-out target-domain text.

Lower is better. It is directly comparable across tokenizers and architectures.

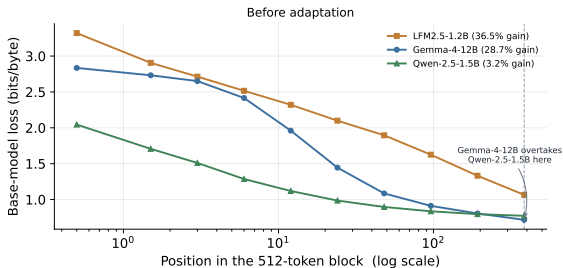
Relative reduction

A diagnostic: how much the released model recalibrates to the corpus's style, formatting, and context pattern.

It explains the trajectory; it does not replace the adapted endpoint.

Token-level evidence makes that interpretation visible.

Context explains the calibration signal



Corpus loss is aggregated over independent 512-token blocks, so nearly all cuts start mid-document with *no preceding context*.

The byte-normalized curves cross.

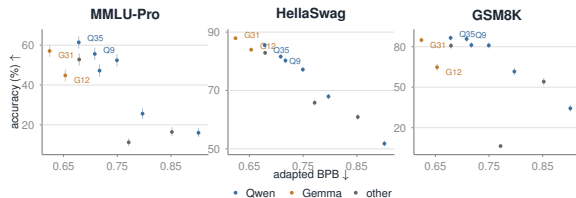
Over the first 8 positions Gemma averages 2.566 bits/byte vs Qwen's 1.490 — but at position 256+ Gemma is **0.717** and Qwen **0.773**.

The crossing survives tokenizer normalization: vocabulary size alone does not explain it. With context, the 12B model shows the stronger steady-state result.

Among the top 200 gain-carrying tokens: ':', ',', '\n\n', 'the', 'and'. Formatting, punctuation, and function words carry 83% of that mass.

Adaptation does not improve the ranking — it rotates it

The zero-shot baseline is the comparison that makes this visible



News BPB	zero-shot	adapted
GSM8K	0.836	0.727
MMLU-Pro	0.855	0.764
HellaSwag	0.718	0.982

Adaptation buys **HellaSwag** agreement and **gives back** GSM8K and MMLU-Pro. Same direction on Reddit and Hacker News.

HellaSwag scores continuations by log-likelihood — the same object BPB measures. That agreement is convergent validity, not proof the ranking got better.

Different tasks reveal different strengths

▲*n*/ ▼*n* = ranks *n* places higher/lower than BPB puts it

Model	Adapted BPB <i>continuation</i>	HellaSwag <i>continuation</i>	MMLU-Pro <i>knowledge</i>	GSM8K <i>arithmetic</i>
Gemma-4-31B	1	1 ●	2 ▼1	3 ▼2
Gemma-4-12B	2	3 ▼1	7 ▼5	7 ▼5
Qwen-3.5-35B-MoE	3	2 ▲1	1 ▲2	1 ▲2
Ministral-3-14B	4	4 ●	4 ●	5 ▼1
Qwen-3.5-9B	5	5 ●	3 ▲2	2 ▲3
Qwen-2.5-7B	6	6 ●	6 ●	4 ▲2
Qwen-3.5-4B	7	7 ●	5 ▲2	6 ▲1
Llama-3.2-1B	8	9 ▼1	11 ▼3	11 ▼3
Qwen-2.5-1.5B	9	8 ▲1	8 ▲1	8 ▲1
LiquidAI-LFM2.5	10	10 ●	10 ●	9 ▲1
Qwen-2.5-0.5B	11	11 ●	9 ▲2	10 ▲1
Ranks identical to BPB	—	7 / 11	3 / 11	0 / 11
Largest displacement	—	1	5	5

HellaSwag preserves the entropy order; MMLU-Pro and GSM8K expose task-specific residuals.

The family pattern explains the shift

GSM8K exposes math-specific ranking differences

Every **Qwen** model is unchanged or ranks **higher** on GSM8K than in the BPB ordering.

Both **Gemma** models and Llama move lower; Ministral moves one place lower.

This is consistent with family-specific math/reasoning emphasis, but the observational comparison does not identify a causal training ingredient.

BPB keeps **language modelling** separate from **arithmetic skill**, so the two measurements can be used together.

The top-two GSM8K gap is only 0.53 points and is not decisive.

One pair, two capabilities

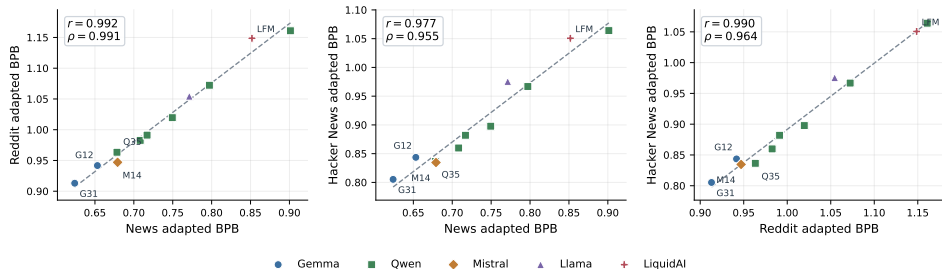
	Qwen-2.5-1.5B	Llama-3.2-1B
Adapted BPB ↓	0.797	0.771
HellaSwag ↑	0.678	0.658
GSM8K ↑	0.611	0.065

Adjacent BPB ranks conceal a **54.6-point** GSM8K gap. HellaSwag differs by only 2.0 points in the other direction.

The preferred model follows the capability the product needs.

Adaptation changes the ranking — then it transfers

Within-domain reordering; cross-domain stability after adaptation



Adaptation changes selection:

zero-shot $\rightarrow$ adapted $\rho = 0.655/0.609/0.682$;
maximum moves are **7/7/5 places**.

Adapted ranks transfer: cross-domain

$\rho = +0.991 / +0.955 / +0.964$; no model moves
more than two places.

Gemma-4-31B moves from zero-shot rank **4/5/6** to adapted rank **1/1/1** on news, Reddit, and Hacker News.

Domain-matched adaptation narrows the GSM8K gap

Same protocol, an independent math-domain corpus instead of news

Adaptation corpus	GSM8K ρ	Pairs correct
<i>none</i> (zero-shot)	0.836	46/55
News	0.727	43/55
Reddit	0.691	42/55
Hacker News	0.791	45/55
Math (arXiv)	0.936	50/55

Three general-text corpora leave GSM8K *at or below* the do-nothing baseline. Only the topically matched corpus beats it — and it is the best selector in the whole study.

Mechanistic: Gemma-4-31B/12B rank 1/2 under news-BPB and fall to 4/7 under math-BPB — exactly the family GSM8K penalizes. Not a sharpened copy of the news order ($\rho = 0.755$); leave-one-out stable ($[0.915, 0.952]$). But at $n=11$ the bootstrap interval on the difference still spans zero — the *pattern* across four corpora is the evidence, not any one contrast.

Not a free lunch: HellaSwag agreement falls $0.982 \rightarrow 0.818$ over the same swap — domain-matched adaptation trades general calibration for task-relevant calibration.

We tried to break it. The ranking held.

Adapted models scored directly on the benchmarks, every baseline re-measured on the same machine.

The test

- Everything so far infers behaviour from *rank movement*. So we adapted the models, kept the adapters, and scored them on the benchmarks directly.
- Every baseline re-measured on the same host, because re-scoring the *identical* model on a different machine moves GSM8K by up to **0.7 points** — wider than many published gaps.

The result

- The ordering was **identical in all 20** model-corpus-benchmark cells — $\rho = 1.00$ everywhere, not one inverted pair.
- Random reordering reproduces that with probability $1/120$ per cell, so the result holds at $p < 0.01$ on the most conservative reading available.
- On the main study, scored as **head-to-head picks** with the uncertainty taken over models: the metric picks the better model above chance in **23 of 24** settings.
- Matched adaptation steered the ranking the predicted way in **93.6%** of 20,000 resamples. We then re-ran it clustering by **publisher** — the harder test, since families are what move — and **not one draw in 20,000** went the other way. The direction is not in question; how *much* one corpus steers is a bigger-cohort question.

The claim we sell is the ranking. It survived a direct attack, and it holds in 23 of 24 settings. Which corpus steers hardest is a bigger-cohort question.

The hardest objection — and the criterion we built to answer it

“You validate an un-preparable metric against the very benchmarks you distrust.”

So we stopped using them

- A criterion from *our own* held-out split: predict a hidden entity or number from 1,200 characters of context. 2,500 items, 11 models.
- **Generate and string-match** — never log-likelihood. A likelihood criterion would agree with BPB by construction and prove nothing.
- Scored two ways, because base models ramble: answer *anywhere* in a short window, or as an exact prefix. Neither was pre-registered, so both ship.
- **Then a second one** from the maths corpus, full 2×2 . Each corpus's ruler wins on its own criterion (**0.982** vs 0.778 news, **0.926** vs 0.870 maths); **none of 16 contrasts is negative** — but every interaction interval contains zero.

Four corpora, each judged on its own criterion

Candidates within $2 \times$ in size — 10 or 11 pairs each

Criterion	adapt	zero-shot	GSM8K	MMLU	size
News	0.909*	0.545	0.182	0.182	0.636
Reddit	1.000 [†]	0.636	0.273	0.273	0.727
Mathematics	0.900	0.900	0.700	0.700	0.800
Hacker News	0.900	0.800	0.500	0.600	1.000 [†]

* the ONE interval in all 56 cells that excludes chance. [†] a perfect

score collapses the bootstrap to $[1, 1]$ — not evidence, excluded.

No benchmark clears chance on any of the four, under either rule

— nor the zero-shot reading, nor model size. At 10 pairs almost nothing is resolvable, so this is a **failure to reject**, not proof the suites are useless.

So we enlarged the cohort — next slide.

Four corpora, criteria no vendor could have prepared for — and at eleven models the one thing that separates from chance where the decision is hard is ours.

We said eleven models was too few to resolve this. So we tested that

Nine more base models, preregistered before a single cell ran. In-band pairs 11 $\rightarrow$ 40.

Selector, 40 in-band pairs	anywhere-in-window	exact prefix
Adapted BPB (ours)	0.900 [0.733, 1.000]	0.850 [0.591, 1.000]
HellaSwag	0.825 [0.593, 0.984]	0.825 [0.545, 0.980]
Zero-shot BPB (zero-shot reading)	0.750 [0.417, 1.000]	0.850 [0.595, 1.000]
Parameter count	0.650 [0.379, 0.841]	0.750 [0.517, 0.932]
MMLU-Pro	0.400 [0.143, 0.688]	0.450 [0.172, 0.744]
GSM8K	0.359 [0.100, 0.657]	0.410 [0.143, 0.718]

We rank first under both rules, and our margin over counting parameters is the only one excluding zero: **+0.250**, [0.074, 0.500]. Seed replication (17 models $\times$ 3 seeds) moves that interval by 0.001 — not a lucky draw.

The half that cuts against us, which we preregistered ourselves to report: HellaSwag **clears chance under both rules** where at 11 models nothing did, and our margin over the best benchmark **does not exclude zero** (+0.075, [0.000, 0.286]).

The idea: stop asking how a new model scores on someone else's exam, and measure how well it predicts your own text. Bits per byte makes that comparable across tokenizers; a fixed short adaptation is how you read it. Every alternative here fails at least one of the three things that makes the idea work — it must be impossible to prepare for, it must hold under both of our scoring rules, and it must reproduce.

What we will not claim

Written down before you ask, because the intervals are in the paper either way

- **“Adapting beats not adapting.”** The paired contrast spans zero under both rules — $+0.150$ $[-0.087, 0.476]$ lenient, $+0.000$ $[-0.278, 0.257]$ strict. Under strict the two readings are *the same number* (0.850). Computed after the fact and reported as exploratory.
- **“BPB beats benchmarks.”** $+0.075$ $[0.000, 0.286]$. Does not exclude zero.
- **“Benchmarks are uninformative.”** HellaSwag clears chance under both rules.
- **“Beats size.”** Lenient only; under strict, size clears chance too.

None of this touches the idea. Measuring a model against your own fresh text is right whether you adapt first or not — what we cannot yet tell you is which of the two readings is sharper, and we say so rather than picking the flattering one. The criterion is generated by *released* weights, so nothing here forecasts a model after *your* fine-tune.

A simple decision rule

What we learned

- One coordinate system valid across tokenizers and architectures.
- Adaptation does not improve the ranking — it **rotates** it toward whatever the adaptation corpus exercises.
- The adapted ranking is stable across three corpora, and a 512-token adapter holds its gain through 2,048-token inference.
- So the adaptation corpus *is* the question you are asking.

How to use it

- Collect *recent* target-domain text — freshness beats any contamination detector.
- **More than $\sim 2\times$ apart in size?** Take the bigger model and stop. Over a $70\times$ range size selects as well as we do, and under the prefix rule it beats us.
- **Otherwise:** adapt each finalist on that text under one budget and rank by adapted BPB. Minutes on one GPU — 4 at 1B, 12 at 8B, 25 at 12B.
- Match the corpus to the capability that decides it.
- **The zero-shot number is a screen, not the selector** — and the reason is mechanical, not statistical. It reads a *convention mismatch* as incapacity. **LFM2.5-1.2B is last of eleven zero-shot** (1.34 bits/byte, behind a 0.5B model at 0.93) and **ahead of that model once adapted** (0.85 vs 0.90) — and on our in-domain criterion it really is ahead. The three models zero-shot under-rates most are exactly the three with the largest adaptation gains. We used to recommend the free number.

Bottom line: one fair ruler, a screen and a selector, and a corpus you choose deliberately — because the corpus decides which capability the ranking tracks.

Thank you

Questions?

Backup slides follow: full benchmark scores, contamination breakdown, adaptation protocol, longer-context transfer.

Backup: why this and not the alternatives

Including where each alternative is the better instrument

Option	Cost each	What it does and does not tell you
Read their scorecard	free	Three things go wrong at once. It does not predict what you need — among candidates within $2\times$, GSM8K picks the better model 0.359 of the time and MMLU-Pro 0.400, worse than a coin flip, neither clearing chance under either rule. You cannot tell what the score means — public exam, undisclosed mixture, so stronger and saw more of it look identical. And the number is not stable — identical weights move 0.7 GSM8K points when only the inference build changes. (HellaSwag does clear chance, 0.825, and we say so.)
Run the suite yourself	GPU-hours	Removes host/harness drift (worth up to 0.7 pts here). Measures the same thing.
Arena / preference	indirect	Rates instructed chat behaviour, not base models. Different question.
Pick the bigger model	free	As good as us over a $70\times$ range (0.855). Collapses to 0.545 within $2\times$.
Label 500 in-house items	human hours	The sharpest objection. But it answers a task you must name in advance, and ± 4.4 pts at $n=500$ is wider than most gaps here.
Build an in-domain cloze set	~ 1 GPU-h	The unlabelled form of the row above — and the instrument we built to check ourselves. If you can build one, do: it measures in-domain ability instead of predicting it. We used it as the criterion, so we cannot compare it against ourselves without circularity.
Fine-tune & test the task	highest	Strictly better evidence when the task is known. Needs labels a new domain lacks.
This protocol	minutes	Text you own, dated after they shipped, so it cannot have been prepared for. 0.818 among comparable-size candidates.

We are not the best answer to every version of the question. We are the right one when the task is not yet a labelled benchmark and the candidates are close in size — which is where the scorecard, measured honestly, stops carrying signal.

Backup: the adaptation results as a table

Lower BPB is better. The reduction is a calibration diagnostic, not plasticity — see slide 9.

Model	Architecture	Params	Zero-shot	Adapted	Reduction
Gemma-4-31B	Dense Transformer	31B	0.752	0.624	17.0%
Gemma-4-12B	Dense Transformer	12B	0.866	0.653	24.5%
Qwen-3.5-35B-MoE	Mixture of Experts	35B	0.707	0.678	4.1%
Ministral-3-14B	Dense Transformer	14B	0.718	0.679	5.4%
Qwen-3.5-9B	Dense Transformer	9B	0.740	0.708	4.4%
Qwen-2.5-7B	Dense Transformer	7B	0.758	0.717	5.4%
Qwen-3.5-4B	Dense Transformer	4B	0.782	0.749	4.2%
Llama-3.2-1B	Dense Transformer	1.2B	0.811	0.771	4.9%
Qwen-2.5-1.5B	Dense Transformer	1.5B	0.833	0.797	4.3%
LiquidAI-LFM2.5	Liquid Neural Net	1.2B	1.340	0.852	36.4%
Qwen-2.5-0.5B	Dense Transformer	0.5B	0.935	0.901	3.6%

Every value reproduces from the marker-masked fixed-protocol files in `results/domain_transfer/news_*.json`.

Backup: full benchmark scores

The ranks and displacements on slide 11 are derived from this table

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K
Gemma-4-31B	0.624	0.578	0.879	0.842
Gemma-4-12B	0.653	0.447	0.843	0.636
Qwen-3.5-35B-MoE	0.678	0.620	0.855	0.865
Ministral-3-14B	0.679	0.541	0.829	0.807
Qwen-3.5-9B	0.708	0.562	0.816	0.860
Qwen-2.5-7B	0.717	0.475	0.803	0.813
Qwen-3.5-4B	0.749	0.525	0.772	0.804
Llama-3.2-1B	0.771	0.117	0.658	0.065
Qwen-2.5-1.5B	0.797	0.250	0.678	0.611
LiquidAI-LFM2.5	0.852	0.155	0.609	0.550
Qwen-2.5-0.5B	0.901	0.167	0.518	0.347

Rank analysis uses the complete 11-model cohort. Harness: *lm-evaluation-harness*, vLLM backend, bf16, greedy decoding, each tokenizer's native BOS. MMLU-Pro 5-shot CoT on a fixed 1,000-question subset; HellaSwag 10-shot length-normalized; GSM8K 5-shot flexible-extract.

Backup: contamination breakdown

$N = 1000$ sampled sequences per model; 80 unique risk-flagged sequences removed

Model	Min-K++ flags	CoDeC flags	DCQ candidates	Preference	Risk flags
google-gemma-4-31B	12	6	20	100%	20
google-gemma-4-12B	11	6	11	100%	11
mistralai-Ministral-3-14B	12	8	13	100%	13
Qwen-Qwen3.5-35B-MoE	6	8	11	100%	11
Qwen-Qwen3.5-9B	12	6	17	100%	17
Qwen-Qwen2.5-7B	10	6	14	100%	14
Qwen-Qwen3.5-4B	9	6	12	100%	12
meta-llama-Llama-3.2-1B	17	0	28	100%	28
Qwen-Qwen2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-Qwen2.5-0.5B	8	2	13	100%	13

These are conservative quarantine signals on a preselected subset, not verified memorization rates. Exact overlap and model-likelihood signals must be interpreted separately.

Backup: adaptation protocol

Data handling

- Google News slice, 8 June 2026: 119,054 articles, ~ 361 MB.
- 80/10/10 train/val/test, seed 42, enforced zero intersection between splits.
- Packed into contiguous 512-token blocks along EOS boundaries.

Scoring

- Corpus-wide cross-entropy over independent 512-token test blocks.
- Context resets at block cuts; injected start markers are context, not targets.
- Model-specific bytes-per-token factor estimated from the test text.

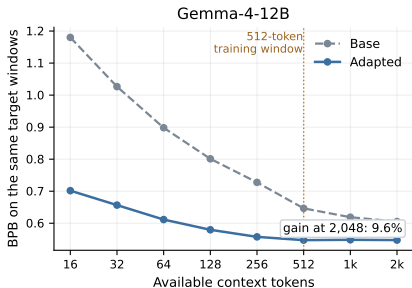
Per-model HPO

- 20-trial Optuna sweep, Successive Halving (top 33% promoted).
- LoRA rank $r \in \{8, 16, 32, 64\}$; α -multiplier $\in \{1, 2, 4\}$; dropout; LR 10^{-6} – 10^{-3} log-scale; effective batch to 128 via gradient accumulation.
- Adapters restricted to the language backbone.
- Winner trained to convergence: cosine schedule to $0.1 \times$ peak LR, gradient clip 1.0.

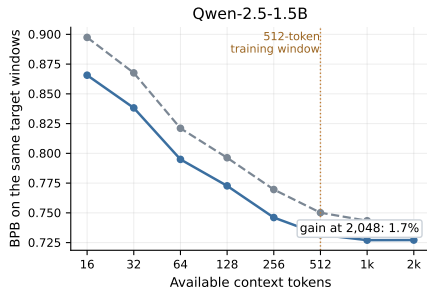
Hardware: A100/H100 multi-node under Slurm.

Backup: 512-token adaptation transfers to longer contexts

Same held-out targets; only the available preceding context changes



Gemma base BPB keeps recovering after 512:
0.6465 → **0.6054** at 2,048 tokens.



Adapted Gemma stays at **0.547**; its advantage remains **9.6%** at four times the training context.

The adapter does not require the inference sequence to stop at 512 tokens.